Week 1 housing forecasting exercise answers

This submission follows all seven numbered exercises in the supplied teaching module. Each exercise includes the requested model, numerical results and interpretation. The three final interpretation questions are answered after Exercise 7. All numbers come from the local housing database and the reproducible Week 1 run.

Data and evaluation dates

The primary raw training window begins in 2015-01. We use 13 monthly validation origins from 2020-12 through 2021-12 and 30 later test origins from 2022-12 through 2025-05. Validation outcomes through the longest horizon are available by the first test origin. The 3-, 6- and 12-month targets are identical across candidate models and eligible counties. Tables below report test errors unless labeled validation.

Geography	Variable	Observed first month	Observed last month	Missing months	Study status
Los Angeles County	zhvi	2000-01	2026-08	0	evaluated
Orange County	zhvi	2000-01	2026-08	0	evaluated
Riverside County	zhvi	2000-01	2026-08	0	evaluated
San Diego County	zhvi	2000-01	2026-08	0	evaluated
California	zhvi	Unavailable	Unavailable	N/A	insufficient coverage for fixed calendar
Los Angeles County	zori_smoothed_sa	2015-01	2026-08	0	evaluated
Los Angeles County	active_inventory	2012-01	2026-05	0	evaluated
Los Angeles County	permits_total	2000-01	2026-06	0	evaluated

We forecast log growth and recover price/rent levels by compounding the predicted changes. Consequently, MAE and RMSE in the price and rent tables are in USD and USD per month, respectively. They measure errors in recovered levels. MAPE compares errors relative to each observed level. These retrospective backtests use revised, provider-smoothed histories. Publication dates and historical release vintages have not been reconstructed.

The introductory level AR(1) estimates P_t = a + b P_{t-1} + e_t; a trending or nonstationary price level can give a high fitted R squared without good forecasts. The growth AR(1) instead models g_t = a + b g_{t-1} + e_t. Recover a horizon-h level as P_hat_(t+h) = P_t × exp[(g_hat_(t+1) + … + g_hat_(t+h)) / 100]. Lag length counts past predictors; horizon counts months ahead. For example, AR(3) can produce a 12-month forecast by recursion. More lags can capture dynamics but estimate more parameters from the available sample. These introductory models and comparisons are included below.

Exercise 1 Lag length comparison

1.1 Estimate AR(1), AR(3), and AR(6) for monthly LA County ZHVI log growth using a consistent sample.

Let P_t be LA County ZHVI and g_t = 100 × [ln(P_t) − ln(P_{t-1})]. For AR(p), the LHS is g_t and the RHS contains an intercept and each separate lag g_{t-1}, …, g_{t-p}, for p equal to 1, 3 and 6. All three models retain six growth observations for initialization so that their fitted response months coincide.

1.2 Record the LHS, every RHS lag, coefficients, number of observations, and training dates.

The following regression snapshot uses information through 2022-12, the first test origin. Training RMSE is in percentage points of monthly log growth. The coefficients table includes every estimated RHS term. The complete origin-by-origin fits are in regressions.csv.

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Growth AR(1)	2015-01	2015-08	2022-12	89	0.4749	0.7412
Growth AR(3)	2015-01	2015-08	2022-12	89	0.3098	0.8899
Growth AR(6)	2015-01	2015-08	2022-12	89	0.2995	0.8971

Model	Raw start	Term	Coefficient
Growth AR(1)	2015-01	intercept	0.080553
Growth AR(1)	2015-01	log_growth_pct_lag_1	0.864224
Growth AR(3)	2015-01	intercept	0.161995
Growth AR(3)	2015-01	log_growth_pct_lag_1	1.486184
Growth AR(3)	2015-01	log_growth_pct_lag_2	-0.705458
Growth AR(3)	2015-01	log_growth_pct_lag_3	-0.036958
Growth AR(6)	2015-01	intercept	0.145308
Growth AR(6)	2015-01	log_growth_pct_lag_1	1.507184
Growth AR(6)	2015-01	log_growth_pct_lag_2	-0.765025
Growth AR(6)	2015-01	log_growth_pct_lag_3	-0.039902
Growth AR(6)	2015-01	log_growth_pct_lag_4	0.287202
Growth AR(6)	2015-01	log_growth_pct_lag_5	-0.446009
Growth AR(6)	2015-01	log_growth_pct_lag_6	0.226295

1.3 Forecast the same later observations and compare 3-, 6-, and 12-month accuracy.

Recovered home-value accuracy is measured in dollars on the same test dates.

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Growth AR(1)	12,228.86	14,938.09	29,022.62	35,224.48	46,040.24	54,855.57
Growth AR(3)	8,688.48	11,761.14	20,828.75	25,910.80	41,226.46	47,851.26
Growth AR(6)	8,270.45	11,589.69	20,149.91	25,467.32	40,337.34	46,322.93

1.4a Which model fits training data best?

Growth AR(6) has the lowest training RMSE in this common-sample snapshot. The three rows above use identical fitted response months, so their training errors are comparable.

1.4b Does it forecast best?

Among these three growth models, the lowest test RMSE comes from Growth AR(6) at all three horizons. The best-fitting model therefore also forecasts best in this comparison. This is a comparison using realized test outcomes. Exercise 3 separately evaluates the specifications that earlier validation data selected.

1.4c Does this change by horizon?

The winning lag length stays the same at all three horizons. The benchmark comparison can still change with horizon. The following table states directly whether the best growth AR beats no change at each horizon.

Horizon	Best growth AR	Growth AR RMSE	No-change RMSE	Growth AR beats no change
3	Growth AR(6)	11,590	18,241	Yes
6	Growth AR(6)	25,467	27,766	Yes
12	Growth AR(6)	46,323	31,587	No

Exercise 2 Window sensitivity

2.1 Keep AR(3), the outcome, and evaluation dates unchanged. Change only the training start.

Hold the LHS, AR(3) lag terms, expanding-window ends and all target dates fixed. Change only the first raw observation to each of the three dates below. The same seven-month growth/lag initialization applies to every start.

2.2 How do the fitted coefficients and forecast errors compare across samples?

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Growth AR(3)	2010-01	2010-08	2022-12	149	0.3079	0.8942
Growth AR(3)	2015-01	2015-08	2022-12	89	0.3098	0.8899
Growth AR(3)	2019-01	2019-08	2022-12	41	0.2322	0.8998

Model	Raw start	Term	Coefficient
Growth AR(3)	2010-01	intercept	0.101827
Growth AR(3)	2010-01	log_growth_pct_lag_1	1.498180
Growth AR(3)	2010-01	log_growth_pct_lag_2	-0.672246
Growth AR(3)	2010-01	log_growth_pct_lag_3	-0.016983
Growth AR(3)	2015-01	intercept	0.161995
Growth AR(3)	2015-01	log_growth_pct_lag_1	1.486184
Growth AR(3)	2015-01	log_growth_pct_lag_2	-0.705458
Growth AR(3)	2015-01	log_growth_pct_lag_3	-0.036958
Growth AR(3)	2019-01	intercept	0.144346
Growth AR(3)	2019-01	log_growth_pct_lag_1	1.609759
Growth AR(3)	2019-01	log_growth_pct_lag_2	-0.900912
Growth AR(3)	2019-01	log_growth_pct_lag_3	0.092455

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Start 2010-01	8,266.22	10,897.42	20,033.29	23,468.57	35,898.57	41,868.88
Start 2015-01	8,688.48	11,761.14	20,828.75	25,910.80	41,226.46	47,851.26
Start 2019-01	9,259.44	12,104.06	23,958.41	27,720.22	44,483.04	48,775.77

Six-month RMSE is $23,469, $25,911, $27,720 as the training start moves through the dates shown. The first-lag coefficients are 1.498, 1.486, 1.610, and the second-lag coefficients are -0.672, -0.705, -0.901. The recent window gives a stronger immediate response and a larger subsequent reversal in this run. Its smaller training error does not translate into the smallest test error.

2.3 Why can periods covering COVID-era dynamics or mortgage-rate changes produce different coefficients?

The latest window starts in 2019-01 and has only 17 fitted responses at the first validation origin. Pandemic-era changes in demand, available supply and prices therefore receive greater weight. Mortgage rates affect borrowing costs and purchasing power, which can change how price growth persists or reverses. An AR model summarizes those relationships through its lag coefficients. Changing the period changes the mixture of market conditions being summarized. These univariate regressions do not identify the separate causal effect of COVID or mortgage rates.

2.4 What is the difference between an expanding and a rolling window?

An expanding window fixes the start and advances the end. A rolling window advances both boundaries to retain a fixed length.

Exercise 3 Expanding window backtest

3.1 Select an initial training period of at least five years.

The initial raw window is 2015-01 through 2020-12, 72 consecutive monthly levels. It exceeds five years and has no missing months. The log difference and common lag initialization reduce the number of fitted responses, as recorded in the regression table.

3.2 Estimate AR(1) and forecast 3, 6, and 12 months ahead.

At the initial 2020-12 origin, retain the monthly levels from 2015-01 through that origin. Construct only backward differences and lags. Estimate each AR model, predict the next month, then feed predicted growth into subsequent unavailable lags. Compound each path from the observed origin level and retain the 3-, 6- and 12-month predictions. For a concrete example, the first test origin gives the following AR(1) forecasts in dollars.

Origin	Horizon	Target month	Forecast	Actual	Forecast minus actual
2022-12	3	2023-03	835,059	806,698	28,361
2022-12	6	2023-06	838,665	811,396	27,269
2022-12	12	2023-12	856,695	871,662	-14,967

3.3 Advance the origin by one month, re-estimate, and repeat until the test period ends.

The test loop starts at 2022-12, advances one calendar month at a time and ends at 2025-05. It produces 30 forecasts for each model and horizon. Each new fit uses the expanded history through its own origin.

3.4 Apply the same procedure to AR(3), AR(6), and a naive benchmark.

AR(3) and AR(6) use the same response months and forecast origins as AR(1). The no-change benchmark predicts the last observed home value at every horizon. All four are scored on identical observed targets.

3.5 Plot predicted versus actual home values and calculate MAE and RMSE separately by horizon.

For each horizon, error = forecast − actual, MAE = mean absolute error, and RMSE = square root of mean squared error. The same procedure covers AR(1/3/6) and no change. The table below contains the requested horizon-specific scores, plus the introductory benchmarks.

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Growth AR(1)	12,228.86	14,938.09	29,022.62	35,224.48	46,040.24	54,855.57
Growth AR(3)	8,688.48	11,761.14	20,828.75	25,910.80	41,226.46	47,851.26
Growth AR(6)	8,270.45	11,589.69	20,149.91	25,467.32	40,337.34	46,322.93
No change	15,192.22	18,240.87	21,830.33	27,766.25	25,687.90	31,586.51
Seasonal naive	30,206.78	34,589.33	28,500.14	33,290.32	25,687.90	31,586.51
Level AR(1)	16,359.78	20,146.64	25,371.51	30,159.69	34,671.27	39,236.12
ARIMA(0,1,1)	13,375.20	15,832.61	21,084.70	26,773.47	25,505.82	31,379.58

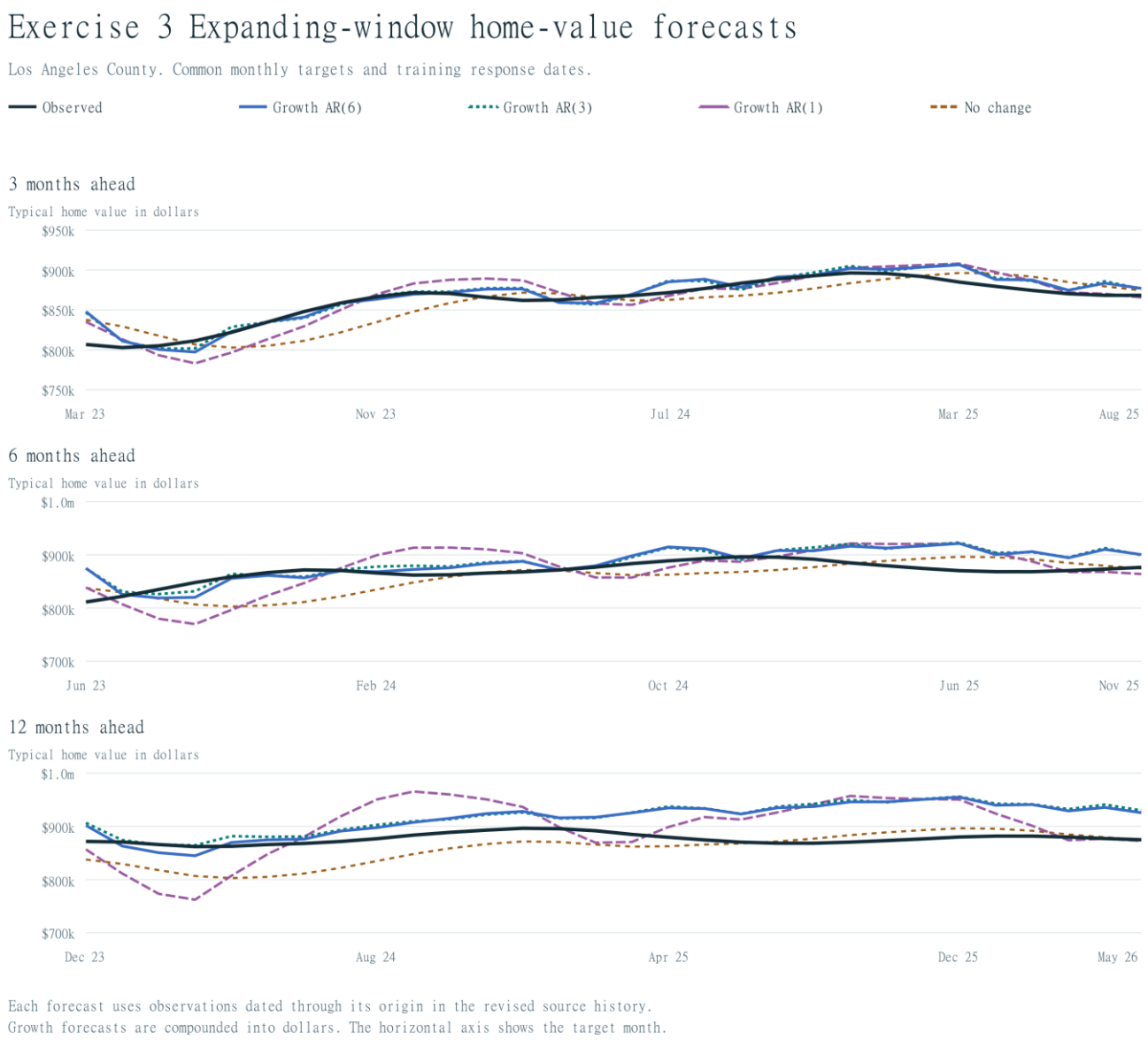
The level AR(1) directly regresses P_t on an intercept and P_{t-1}. The introductory ARIMA(0,1,1) uses log prices, no drift, and a moving-average term in the differenced series. The seasonal benchmark repeats the corresponding month one year earlier.

The benchmark fit snapshot below also uses the 2022-12 origin. Level AR(1) training RMSE is in dollars. ARIMA innovation RMSE is in percentage points of monthly log growth, and OLS R squared does not apply to that conditional innovation fit.

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Level AR(1)	2015-01	2015-04	2022-12	93	5587.3974	0.9977
ARIMA(0,1,1)	2015-01	2015-02	2022-12	95	0.6224	N/A

Model	Raw start	Term	Coefficient
Level AR(1)	2015-01	intercept	-1299.260338
Level AR(1)	2015-01	level_lag_1	1.008247
ARIMA(0,1,1)	2015-01	innovation_lag_1	0.889753

Observed values and each lag model at three horizons



3.6 Keep an early validation period for choosing specifications and reserve a later independent test period.

Validation origins run from 2020-12 through 2021-12. Test origins run from 2022-12 through 2025-05. For independent evaluation, choose a model and training start using the minimum validation RMSE for that horizon, then assess that choice on the later test origins.

Horizon	Validation choice	Start	Validation RMSE	Test MAE	Test RMSE	No-change test RMSE
3	Level AR(1)	2015-01	7,819	16,360	20,147	18,241
6	Level AR(1)	2015-01	13,308	25,372	30,160	27,766
12	Growth AR(1)	2015-01	31,351	46,040	54,856	31,587

The validation-selected specifications have equal or larger test RMSE than no change at all three horizons. Good fitted values and even good validation performance do not guarantee performance in a later market period. All comparisons use matching dates, and adjacent multi-month errors overlap.

3.7 How is leakage avoided, and what limitation remains?

At each origin, slice the observed history before creating differences or lags and estimating coefficients. Recursion uses predicted future growth whenever a lag extends beyond the origin. Actual future values enter only the subsequent error calculation. No full-sample standardization or additional smoothing is applied. Provider revisions and smoothing are already embedded in the published histories, so these results are explicitly retrospective rather than real-time reconstructions of past information sets.

Exercise 4 Rent growth forecasts

4.1 Estimate AR(1) for rent growth and state the LHS and RHS.

Use the active all-homes-plus-multifamily ZORI, which Zillow smooths and seasonally adjusts. Let r_t = 100 × [ln(R_t) − ln(R_{t-1})]. The LHS is r_t and the RHS is an intercept plus r_{t-1}. Recover forecast rent levels by compounding predicted growth from the origin rent.

4.2 Repeat with three and six lags.

AR(3) adds r_{t-2} and r_{t-3}; AR(6) additionally includes r_{t-4}, r_{t-5}, and r_{t-6}, each with its own coefficient. All three fits use the same response months. The next tables give the fitted equations and test MAE/RMSE in dollars per month.

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Growth AR(1)	2015-01	2015-08	2022-12	89	0.1529	0.7909
Growth AR(3)	2015-01	2015-08	2022-12	89	0.1414	0.8213
Growth AR(6)	2015-01	2015-08	2022-12	89	0.1340	0.8394

Model	Raw start	Term	Coefficient
Growth AR(1)	2015-01	intercept	0.044194
Growth AR(1)	2015-01	log_growth_pct_lag_1	0.891254
Growth AR(3)	2015-01	intercept	0.057681
Growth AR(3)	2015-01	log_growth_pct_lag_1	0.837147
Growth AR(3)	2015-01	log_growth_pct_lag_2	0.398861
Growth AR(3)	2015-01	log_growth_pct_lag_3	-0.371603
Growth AR(6)	2015-01	intercept	0.059472
Growth AR(6)	2015-01	log_growth_pct_lag_1	0.915880
Growth AR(6)	2015-01	log_growth_pct_lag_2	0.339349
Growth AR(6)	2015-01	log_growth_pct_lag_3	-0.652550
Growth AR(6)	2015-01	log_growth_pct_lag_4	0.347820
Growth AR(6)	2015-01	log_growth_pct_lag_5	0.133767
Growth AR(6)	2015-01	log_growth_pct_lag_6	-0.223141

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Growth AR(1)	10.60	14.46	18.28	25.77	39.09	52.74
Growth AR(3)	12.39	16.61	22.58	31.16	49.43	62.26
Growth AR(6)	11.36	15.13	20.65	29.49	52.32	64.57
No change	14.83	17.50	28.22	31.06	54.70	58.51

The best rent lag specification by test RMSE is Growth AR(1) at all three horizons.

4.3 Is rent growth more or less predictable than home-value growth over the same dates?

The following percentage errors compare rents and home values on identical dates.

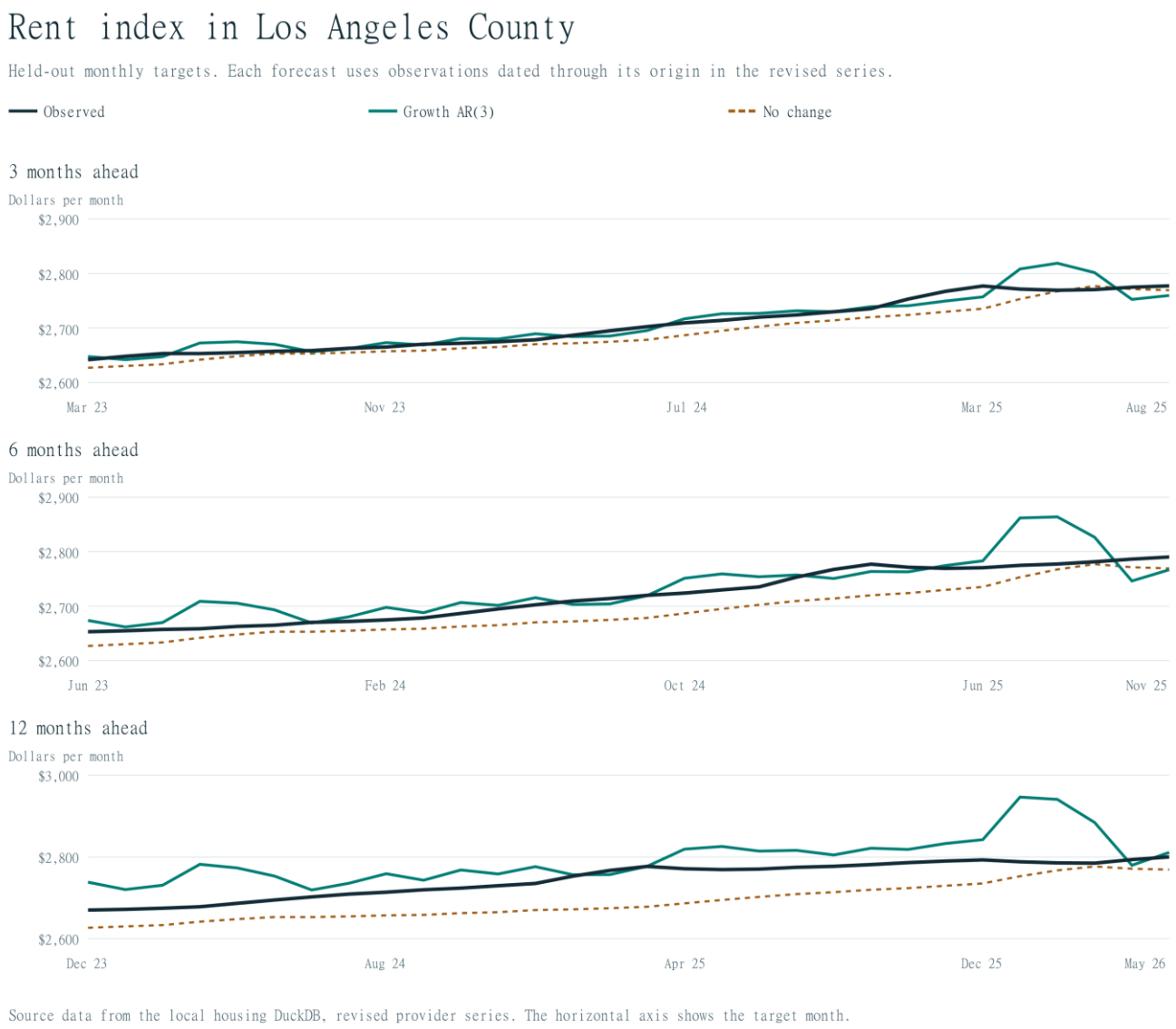
Model	Horizon	Home-value MAPE	Rent MAPE
Growth AR(1)	3	1.44%	0.39%
Growth AR(1)	6	3.36%	0.67%
Growth AR(1)	12	5.26%	1.42%
Growth AR(3)	3	1.01%	0.45%
Growth AR(3)	6	2.41%	0.83%
Growth AR(3)	12	4.70%	1.80%
Growth AR(6)	3	0.97%	0.42%
Growth AR(6)	6	2.33%	0.76%
Growth AR(6)	12	4.60%	1.91%

At six months, growth AR(3) has rent MAPE of 0.83% and price MAPE of 2.41%. Rent has lower percentage forecast errors in every matched model/horizon comparison in the table, so it is more predictable on this recovered-level metric. This is a comparison of growth-model forecasts recovered into published index levels; it does not equate low level MAPE with a high explained share of monthly growth variation.

4.4 Are the rent and price indexes seasonally adjusted or smoothed?

Both active ZORI and ZHVI are provider-smoothed and seasonally adjusted. No additional smoothing or seasonal adjustment is applied. The archived non-seasonally-adjusted ZORI vintage is excluded. Provider processing can affect apparent persistence and predictability, and revised histories limit real-time interpretations.

Rent forecasts and observed values



Source data from the local housing DuckDB, revised provider series. The horizontal axis shows the target month.

Exercise 5 Housing permit activity

5.1 Construct a trailing three-month moving average of permits.

Permits measure housing units authorized during each month. Construct M_t as ($permits_t$ + $permits_{t-1}$ + $permits_{t-2}$) / 3. The average uses the current and two prior months only; it never includes a future observation. The first two raw months initialize the average.

5.2 Regress the current moving average on its own first lag.

The fitted equation is M_t = a + b M_{t-1} + e_t. Its LHS is the current three-month mean; the RHS contains an intercept and the preceding month's three-month mean. The tables also show the raw-permit AR(1), with current monthly permits on the LHS and last month's permits on the RHS.

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Level AR(1)	2015-01	2015-04	2022-12	93	595.4199	0.0065
Mean AR(1)	2015-01	2015-04	2022-12	93	246.4189	0.3967

Model	Raw start	Term	Coefficient
Level AR(1)	2015-01	intercept	1990.774302
Level AR(1)	2015-01	level_lag_1	-0.078742
Mean AR(1)	2015-01	intercept	695.820634
Mean AR(1)	2015-01	mean3_lag_1	0.623226

5.3 Compare forecasts with a model estimated directly on raw monthly permits.

Raw monthly target. MAE and RMSE are in permitted units per month.

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Level AR(1)	348.98	445.98	341.65	444.52	420.75	714.57
No change	421.07	524.59	454.27	574.95	539.53	823.53
Seasonal naive	407.77	489.71	410.73	491.59	539.53	823.53

Common trailing-mean target. Every row forecasts the same observed three-month mean. For the raw AR(1) comparison, average its predicted monthly levels, using observed components whenever those components fall on or before the origin.

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Mean AR(1)	208.99	254.41	242.81	277.40	346.81	490.55
Raw AR(1)	236.67	275.13	240.75	275.68	345.49	488.60
No change	224.80	303.23	309.32	397.08	448.60	645.74
Seasonal naive	273.39	357.26	278.64	362.50	448.60	645.74

At six months, direct mean AR(1) has RMSE 277.4, raw AR(1) followed by averaging has 275.7, all on the common mean target. The direct mean model has lower RMSE at 3 months. Smoothing therefore does not automatically produce the better forecasting method once the target is held fixed.

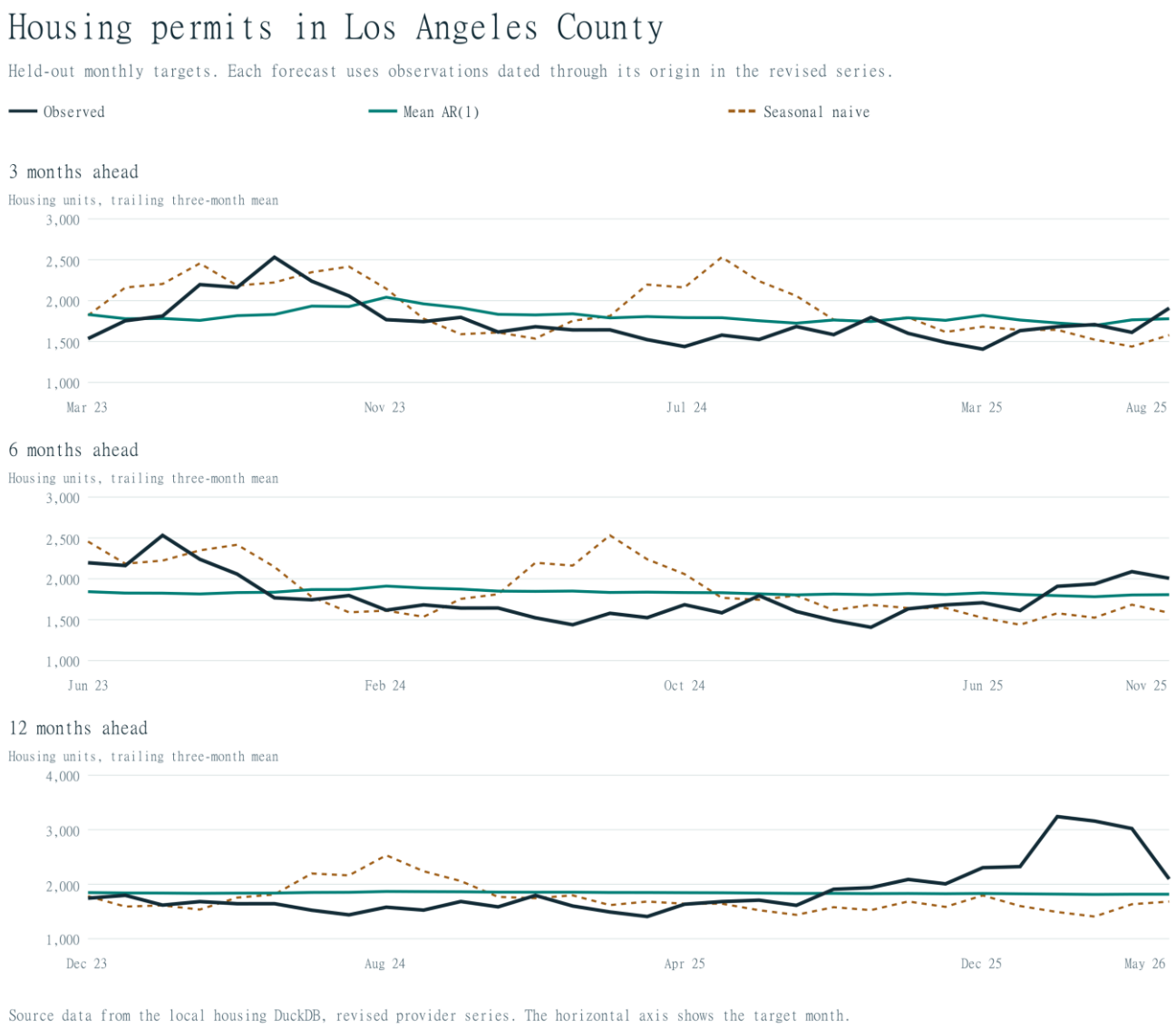
5.4 Compare with a seasonal-naive forecast.

Seasonal naive repeats the value from twelve months before the target. Its mean-target variant averages the three corresponding year-earlier monthly values. At six months its raw-target RMSE is 491.6, versus 444.5 for raw AR(1); its mean-target RMSE is 362.5, versus 277.4 for direct mean AR(1). The horizon-specific tables above retain the full comparison.

5.5 Why does overlapping smoothing create persistence, and how should forecast errors be compared?

Adjacent averages share two monthly observations, which creates persistence mechanically. For example, M_t and M_{t-1} both contain $permits_{t-1}$ and $permits_{t-2}$, even if raw monthly flows were independent. A smoother target also has less variation. Compare competing methods against the same observed target; a lower RMSE for means than for raw flows is not by itself evidence of improved forecasting. The raw and averaged targets therefore have separate tables. Census flags provider imputation in most monthly estimates, and monthly coverage expanded in 2022.

Permits forecasts on a common trailing-mean target



Source data from the local housing DuckDB, revised provider series. The horizontal axis shows the target month.

Exercise 6 Housing inventory

6.1 Verify whether a suitable inventory series is available.

Use Redfin county active inventory, an unadjusted monthly stock. The level models regress actual county inventory; no ZIP active-listings measure is relabeled as inventory. The data inventory at the start of this report records its observed coverage and confirms a complete monthly study interval.

6.2 Estimate AR(1) and AR(3) in inventory levels.

AR(1) has $inventory_t$ on the LHS and an intercept plus $inventory_{t-1}$ on the RHS. AR(3) adds $inventory_{t-2}$ and $inventory_{t-3}$, each separately. Both use the same fitted response months within the level comparison.

6.3 Consider log changes if the inventory observations are positive.

All available inventory observations are positive. Level-regression training RMSE is measured in homes, and growth-regression training RMSE in percentage points.

Define q_t = 100 × [ln($inventory_t$) − ln($inventory_{t-1}$)]. The growth specifications regress q_t on an intercept and its first lag or its first three lags. Forecast growth is compounded back into homes, so every row in the test table can be compared on the same level target.

Model	Raw start	First fitted month	Last fitted month	N	Training RMSE	R squared
Level AR(1)	2015-01	2015-04	2022-12	93	1119.0147	0.8326
Level AR(3)	2015-01	2015-04	2022-12	93	972.1093	0.8737
Growth AR(1)	2015-01	2015-08	2022-12	89	7.3864	0.1566
Growth AR(3)	2015-01	2015-08	2022-12	89	7.3320	0.1690

Model	Raw start	Term	Coefficient
Level AR(1)	2015-01	intercept	1106.066081
Level AR(1)	2015-01	level_lag_1	0.927616
Level AR(3)	2015-01	intercept	1785.654142
Level AR(3)	2015-01	level_lag_1	1.407950
Level AR(3)	2015-01	level_lag_2	-0.551196
Level AR(3)	2015-01	level_lag_3	0.029065
Growth AR(1)	2015-01	intercept	-0.427468
Growth AR(1)	2015-01	log_growth_pct_lag_1	0.409131
Growth AR(3)	2015-01	intercept	-0.434843
Growth AR(3)	2015-01	log_growth_pct_lag_1	0.454884
Growth AR(3)	2015-01	log_growth_pct_lag_2	-0.117960
Growth AR(3)	2015-01	log_growth_pct_lag_3	-0.014314

Specification	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Level AR(1)	1,528.40	1,917.90	2,051.78	2,548.54	2,173.34	2,599.07
Level AR(3)	1,620.74	1,973.80	2,408.69	2,776.82	2,124.24	2,689.29
Growth AR(1)	1,570.80	1,991.22	2,351.95	3,093.94	3,161.39	3,569.32
Growth AR(3)	1,602.66	2,005.85	2,285.76	3,020.66	3,152.59	3,617.67
No change	1,589.47	1,979.00	2,143.37	2,824.03	2,744.13	3,248.19
Seasonal naive	3,126.27	3,490.20	3,215.90	3,522.12	2,744.13	3,248.19

Between the two level models, the lower test RMSE comes from Level AR(1) at all three horizons. Across all four fitted inventory models, the best RMSE comes from Level AR(1) at all three horizons.

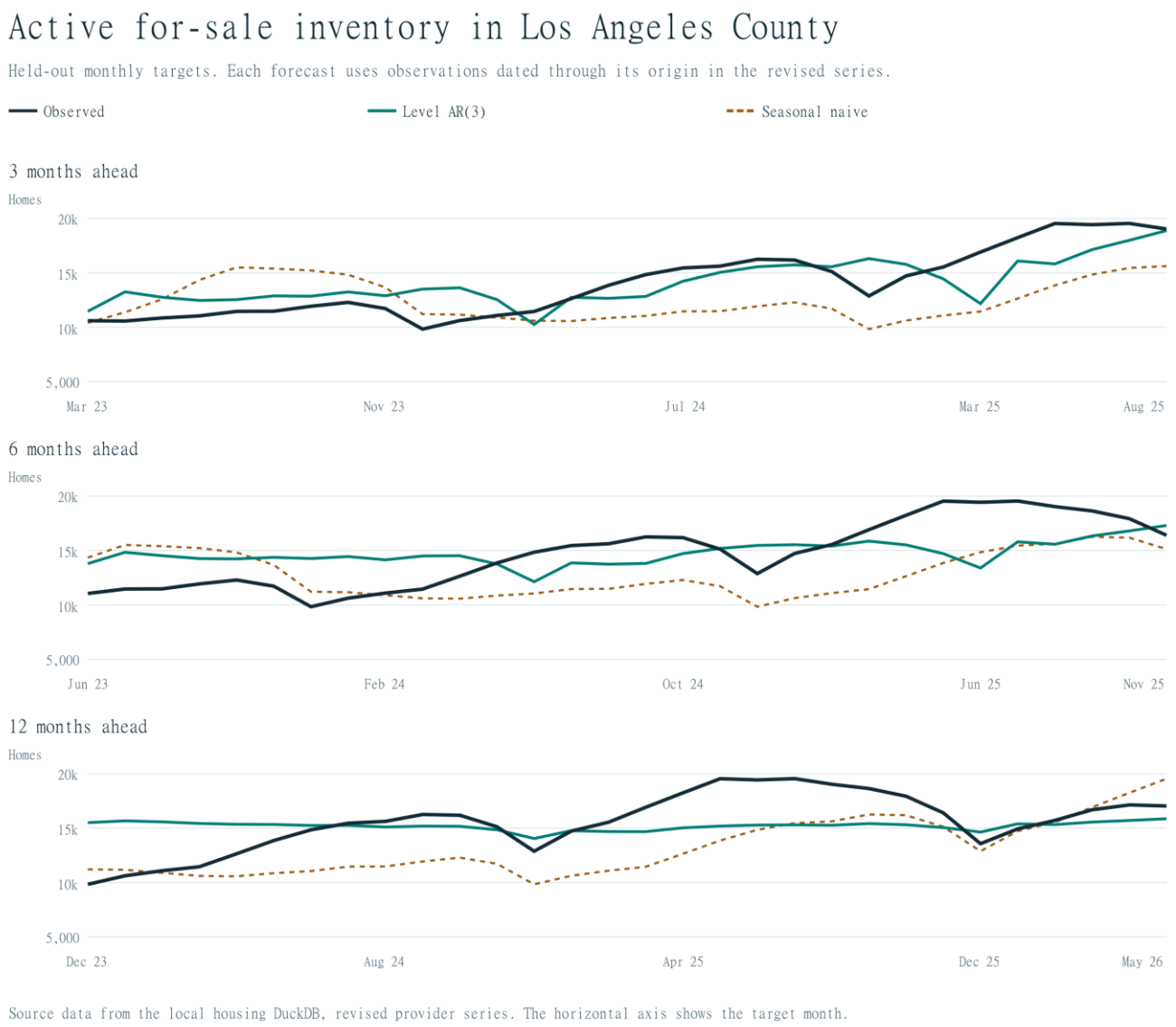
6.4 Compare with a seasonal-naive benchmark.

The seasonal benchmark uses inventory in the target calendar month one year earlier. At six months it has RMSE 3,522 homes, versus 2,549 for level AR(1). The table reports its 3- and 12-month errors as well. Strong seasonality alone does not guarantee that a year-earlier observation forecasts well when the overall inventory level changes.

6.5 Does inventory have stronger seasonality than prices?

Calendar-month means explain 87.4% of monthly inventory log-change variation versus 5.5% for prices over matching dates through the first test origin. The inventory series has a stronger seasonal pattern in this comparison, while ZHVI is already smoothed and seasonally adjusted by its provider.

Inventory forecasts and seasonal benchmark



Source data from the local housing DuckDB, revised provider series. The horizontal axis shows the target month.

Exercise 7 California county comparison

7.1 Verify comparable data for at least two other counties and estimate AR(1) and AR(3).

Repeat growth AR(1) and AR(3) for Orange, Riverside and San Diego in addition to LA. All four have complete ZHVI histories for the study interval, as recorded in the inventory table. The county accuracy table below reports the results, and the full county coefficients and sample dates are retained in regressions.csv.

7.2 Keep units, transformations, lag definitions, horizons, and evaluation dates comparable.

All use the same ZHVI definition in dollars, log-growth transformation, raw training start, six-growth-month initialization, forecast horizons and test dates.

County	Model	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Los Angeles	Growth AR(1)	12,229	14,938	29,023	35,224	46,040	54,856
Los Angeles	Growth AR(3)	8,688	11,761	20,829	25,911	41,226	47,851
Los Angeles	No change	15,192	18,241	21,830	27,766	25,688	31,587
Orange	Growth AR(1)	12,467	14,983	28,574	35,234	55,255	65,377

County	Model	3m MAE	3m RMSE	6m MAE	6m RMSE	12m MAE	12m RMSE
Orange	Growth AR(3)	9,320	11,621	23,515	27,428	45,860	51,647
Orange	No change	22,221	25,308	39,882	45,899	62,197	77,460
Riverside	Growth AR(1)	3,699	4,795	10,350	13,529	27,164	32,251
Riverside	Growth AR(3)	3,275	3,989	10,253	12,075	30,214	33,230
Riverside	No change	6,488	7,301	11,393	12,728	17,853	20,320
San Diego	Growth AR(1)	9,453	11,357	23,976	28,765	52,543	60,569
San Diego	Growth AR(3)	7,159	8,795	21,917	25,857	47,725	59,601
San Diego	No change	13,857	17,486	25,886	31,576	42,295	51,184

7.3 Does the same lag length work best in each county?

County	Lower RMSE at 3m	At 6m	At 12m	6m AR(3) MAPE	Growth SD in pp
Los Angeles	Growth AR(3)	Growth AR(3)	Growth AR(3)	2.41%	0.647
Orange	Growth AR(3)	Growth AR(3)	Growth AR(3)	2.12%	0.601
Riverside	Growth AR(3)	Growth AR(3)	Growth AR(1)	1.69%	0.370
San Diego	Growth AR(3)	Growth AR(3)	Growth AR(3)	2.36%	0.596

The preferred lag depends on both county and horizon. At 3 months: Los Angeles favors Growth AR(3), Orange favors Growth AR(3), Riverside favors Growth AR(3), San Diego favors Growth AR(3). At 6 months: Los Angeles favors Growth AR(3), Orange favors Growth AR(3), Riverside favors Growth AR(3), San Diego favors Growth AR(3). At 12 months: Los Angeles favors Growth AR(3), Orange favors Growth AR(3), Riverside favors Growth AR(1), San Diego favors Growth AR(3).

7.4 Are forecast errors larger in more volatile markets?

Los Angeles has the highest growth standard deviation and Los Angeles the highest six-month AR(3) MAPE. Riverside has the lowest volatility and Riverside the lowest MAPE. The full ranking differs, so greater volatility does not consistently imply a larger error. The six-month MAPE and growth standard deviation use matching target months. MAPE removes the immediate effect of different dollar price levels. The relationship is partly consistent at the extremes, but four counties do not establish a general statistical or causal relationship.

7.5 If statewide data are available, estimate a comparable statewide model.

Statewide ZHVI has no observations in this database, so a statewide model could not be estimated. The optional state comparison is therefore governed by actual source availability and the same fixed-date eligibility checks as the county models.

7.6 Can an unweighted average of the available counties be treated as a statewide index?

No. These four counties do not cover California, and equal county weights ignore differences in housing stock and the provider's index construction. No county average is substituted for an official statewide series.

Final interpretation questions

1. How much of future housing-market behavior can be predicted from the variable's own past?

Past levels and growth contain useful short-horizon information, but the advantage over a simple benchmark varies. For LA home values, growth AR(3) has six-month test RMSE of $25,911, compared with $27,766 for carrying the last observed value forward. The validation-selected specifications have equal or larger test RMSE than no change at all three horizons. No change itself uses past information. The comparison measures how much estimating additional dynamics improves the forecasts in this sample. Rent AR(3) had six-month MAPE 0.83%, compared with 2.41% for home values. These are sample-specific forecast errors, not estimates of a universal share of future housing behavior that is predictable.

2. How sensitive are the results to lag length, training period, and prediction horizon?

The LA growth-lag comparison favors Growth AR(6) at all three horizons. Keeping AR(3) fixed, alternative starts yield six-month RMSEs of $23,469, $25,911, $27,720. For the main AR(3), RMSE is $11,761 at three months and $47,851 at twelve months. Longer forecasts accumulate uncertainty and expose the model to changes in market dynamics. The short recent window changes the coefficients and sacrifices observations. Training fit alone is insufficient to choose a forecast model, and an earlier validation winner can lose on the later test.

3. Do the same simple models work across home prices, rents, permits, inventory, and California counties?

Among the tested growth models, LA price accuracy favors Growth AR(6) at all three horizons, while rent accuracy favors Growth AR(1) at all three horizons. The better inventory level model is Level AR(1) at all three horizons. Permit models must be compared on a common raw or trailing-mean target because averaging changes both persistence and the outcome's volatility. At twelve months, the lower-error growth model is Los Angeles Growth AR(3), Orange Growth AR(3), Riverside Growth AR(1), San Diego Growth AR(3). Seasonality, provider adjustments, volatility and the estimation period all matter. These results support separate validation for each outcome and horizon.

Full numerical evidence is retained in accuracy.csv, regressions.csv, forecasts.csv, selections.csv and transformation_log.csv.